



Saba Shahsavari, Alessa Mayer

Pattern Extraction From Folktales

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Supervised by: Fabian M. Suchanek

Abstract

This project focuses on extracting recurring narrative patterns, known as motifs, from fairy tales using natural language processing techniques. Using a dataset of 1518 folktales classified by Aarne-Thompson-Uther (ATU) indices, we applied automated methods to summarize the stories, extract motifs as sentences, and cluster them to identify common patterns such as "A prince marries a poor girl" across different tale types. By calculating pairwise similarities between motif embeddings, we quantified the relationships between various ATU indices, revealing how different tale types are interconnected. Our findings highlight the most prevalent motifs in the dataset and demonstrate significant narrative similarities across stories.

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Chapter 1

Introduction

1.1 Problem statement and motivation

Stories, from fairy tales to myths, encode cultural values and narrative structures that persist across regions and eras. Common patterns, such as a prince marrying a poor girl despite social barriers (as seen in tales like Cinderella, Ye Xian, or The Rough-Faced Girl) or a dream-guided quest for treasure, suggest shared storytelling archetypes. However, manually identifying these patterns across large, diverse corpora is labor-intensive, and computational approaches must address challenges like motif synonymy (e.g., "Dream prompts a journey" vs. "Dream conveys important information"), narrative variety within tale types, and geographic context.

This project aims to automate the extraction of narrative motifs to explore how these patterns cluster, vary by region, and contribute to the structure of tale types, as categorized in the Aarne-Thompson-Uther (ATU) index. Using natural language processing techniques on a corpus of 1518 folktales, we developed a method to reconstruct ATU groupings based on story motifs. This framework allows us to investigate key questions such as: Which motifs are shared across different story types? and How does regional provenance influence narrative form?

These insights can deepen our understanding of shared human storytelling traditions and their cultural adaptations.

1.2 Related Works

The study of fairy tales has long recognized the crucial role of recurring narrative elements, in understanding their structure and cultural significance. Early efforts in fairy tale classification, pioneered by Antti Aarne 1910 in his "Verzeichnis der Märchentypen" and subsequently expanded by Stith Thompson 1961 in "The Types of Folktale", laid the groundwork for identifying and categorizing these patterns. Thompson's monumental "Motif-Index of Folk-Literature" (Thompson 1955–1958) represents a comprehensive attempt to catalogue the individual narrative elements that constitute these tales, ranging from magical objects and supernatural adversaries to specific tasks and character archetypes. This index serves as an invaluable tool for researchers seeking to trace the distribution and variation of specific motifs across different cultures and time periods.

Building upon this foundational work, Hans-Jörg Uther's revised and expanded "Types of International Folktales" (Uther 2011) (ATU) further refines the classification system, integrating a more nuanced understanding of how motifs contribute to the identification of tale types. While the structuralist approach of Vladimir Propp in *Morphology of the Folktale* (Propp 1968) focused on the sequence of functions within a narrative, the analysis of motifs complements this by highlighting the specific content and recurring details that shape these functions.

Contemporary scholarship, as overviewed in works like Pöge-Alder's "Märchenforschung. Theorien, Methoden, Interpretationen" (Pöge-Alder 2007), continues to recognize the importance of motifs in understanding both the common threads and the unique adaptations of fairy tales. The examination of motifs al-

allows for a deeper appreciation of the underlying cultural values, as well as the creative genius of storytellers across generations.

Other important works on folktales include the contributions of early collectors and authors like Charles Perrault with "Contes de ma mère l'Oye" (Perrault 1697) and Jeanne-Marie Leprince de Beaumont with "Magasin des enfants" (Leprince de Beaumont 1798), whose works have contributed significantly to the enduring canon of fairy tales and their characteristic motifs. The systematic study of motifs, therefore, remains central to unraveling the complex tapestry of the fairy tale tradition.

The earlier mentioned findings and classification of Aarne, Thompson and Uther give the fundament of our results. Hence, in the following section we are introducing the ATU Index in more depth.

1.2.1 Aarne-Thompson-Uther Tale type (ATU Index)

To study and organize the wide variety of folktales, scholars have developed several classification systems. One of the most widely used is the Aarne-Thompson-Uther (ATU) tale type index. This system includes over 2,400 unique entries, arranged in a hierarchical structure that groups stories into types and subtypes based on recurring plot patterns and narrative elements.

Although storytelling offers seemingly infinite creative possibilities, folktales across cultures often rely on a limited set of recurring structures and motifs. The ATU classification is built upon these common narrative elements, grouping different variants of a story under a single category (Dundes 1997). As such, it serves as a valuable tool in the field of folkloristics for analyzing the shared structures that underpin diverse storytelling (Anonymous 2023).

The ATU indices, along with category and subcategory names from the ATU hierarchy, is provided in Table A.1 in the appendix (Multilingual Folk Tale Database 2025).

1.2.2 Motifs and Narrative Patterns

In folkloristics and narratology, motifs are recognized as recurring thematic elements or narrative units that appear across different stories and cultures. The combination of several motifs forms patterns, and allows researchers to describe, analyze and interpret the differences between one culture and another.

Motifs are deeply embedded in cultural narratives and often carry implicit meanings that are understood within specific cultural or national groups. Acharya et al. 2024 describe motifs as distinctive, recurring, widely used idiom-like words or phrases, often originating in folklore and usually strongly anchored to a particular cultural or national group. These motifs function as communicative devices that convey complex cultural associations succinctly.

1.2.3 Previous Work on Automated Analysis of Folktales

While the computational analysis of folk narratives has gained attention in recent years, direct approaches to narrative motif extraction remain underexplored. Existing research primarily focuses on higher-level classification tasks such as story type or genre identification, which, although related, do not entail the extraction of motifs as standalone narrative elements. For instance, the work of Nguyen, Trieschnigg, and Theune 2013 addresses the automatic classification of folktales into story types according to schemas such as the earlier mentioned Aarne-Thompson-Uther (ATU) index. Their method employs a learning-to-rank framework using a combination of information retrieval techniques, lexical similarity measures (e.g., TF-IDF, Jaccard), and structural features such as subject-verb-object (SVO) triplets. Although motifs are acknowledged as part of what defines a story type, the system does not attempt to extract them individually. Instead, motifs contribute implicitly to the overall textual and event-level similarity that informs the classification process.

Similarly, Nguyen, Trieschnigg, Meder, et al. 2012 focus on the automatic clas-

sification of folk narrative genres, such as fairy tales, urban legends, and jokes. The authors mention the use of lexical (e.g., character n-grams), structural (e.g., sentence length, punctuation), and metadata features (e.g., story keywords and summaries) to classify the stories. Though the approach is effective for genre prediction, it does not model or extract motifs as discrete entities. All in all, these classification-based approaches suggest that motifs play a conceptual role in defining narrative structure but are not computationally modeled or extracted in isolation. Thus, while research in story type classification and genre recognition informs the broader context, there is a significant lack of automated motif extraction in the current research.

1.2.4 Automated Detection of Tropes in Short Texts

Flaccavento et al. 2025 propose a method for detecting tropes in short social media posts using supervised learning. While their focus is thus on contemporary discourse and predefined trope categories, our work targets richer, culturally grounded motifs in folktales, which are extracted without prior labels. Unlike their classification-based approach, we identify motifs from scratch and analyze them semantically to uncover narrative structure and inter-story relationships.

1.3 Dataset

Among the publicly available ATU-annotated folktale collections, the most promising source was Ashliman’s Folktexts Collection. This website (Ashliman 2021) , maintained by D.L. Ashliman since 1996, is one of the most comprehensive and extensively annotated collections of folktales online. What makes this collection particularly valuable for computational analysis is its consistent annotation with ATU tale type identifiers, which allows for large-scale classification and pattern analysis.

However, using this data directly from the website requires extensive preprocessing, including web scraping, text extraction, and cleaning. To avoid this overhead, we opted to use a preprocessed version of the dataset provided by Hagedorn and Darányi 2022, who carefully extracted and annotated the tales from Ashliman’s website and made the dataset publicly available.

This version, referred to as the AFT dataset (The Annotated Folk Tales), contains tales from 214 pages of the Ashliman’s site which contains folktales that have been marked as characteristic of a specific tale type from the ATU. The dataset comprises 1518 tales, each represented as a row in a structured dataframe. Each row contains eight columns, briefly described below:

- **atu_id**: The ATU tale type identifier classifying the story.
- **talet_title**: The title of the folktale.
- **provenance**: The origin of the tale, which may refer to the individual who recorded it (e.g., Giambattista Basile), the region it came from (e.g., North Africa), or the collection in which it was originally published (e.g., Kathasaritsagara).
- **notes**: Additional notes relevant to the tale.
- **source**: The bibliographic citation for the tale’s original source.
- **text**: The full body of the tale.
- **data_source**: The dataset’s origin. At the time of writing, all tales are from Ashliman’s Folktexts, though this may expand in the future.
- **date_obtained**: The date the dataset was downloaded and compiled.

The 1518 tales in the dataset vary significantly in length, ranging from as short as 10 tokens to as long as 12406 tokens, with an average length of 979.1 tokens per story. The dataset includes tales from 182 distinct ATU types, with an average of 8.3 tales per type, and a distribution range from 1 to 31 tales per category. To

ensure sufficient data for meaningful pattern discovery, we excluded ATU types with fewer than five stories. After this filtering step, the final dataset used for analysis consists of 1331 stories across 133 distinct ATU types.

Chapter 2

Approach 1

2.1 Methods and Experiments

In this section, we describe the methodology used to identify narrative motifs from a large corpus of folktales. We begin by summarizing the stories using large language models, then explore different strategies for extracting recurring patterns from these summaries. We discuss the models selected, challenges encountered (such as input length limits), and the techniques developed to overcome them. Finally, we detail the motif extraction process and how we refined prompts and outputs to ensure accuracy and consistency.

2.1.1 Story Summarization and Pattern Extraction

Our initial approach involved summarizing the corpus of approximately 1400 folktales using Large Language Models to identify recurring narrative patterns or motifs. We utilized LangChain, a framework for building LLM-powered applications, to streamline the process. Given the diversity of available LLMs, we selected models based on specific criteria: they needed to be freely accessible, excel at instruction-following, perform well on summarization tasks, and be computationally efficient for processing a large dataset. The evaluated models were:

- meta-llama/Meta-Llama-3-8B-Instruct
- mistralai/Mistral-7B-Instruct-v0.2
- facebook/bart-large-xsum

The first two models are instruction-tuned transformer models known for strong zero-shot and few-shot performance. The third model, facebook/bart-large-xsum, is fine-tuned specifically for extreme summarization using the XSum dataset (Narayan et al. 2018).

Input Length Limitations

A major challenge we faced when using large language models to summarize full-length folktales was the limitation imposed by their maximum input token capacity. Since most stories in our dataset are lengthy and rich in narrative detail, fitting them within the allowable token limits required careful strategy.

Each LLM has a fixed maximum context length—the total number of tokens it can process in a single input, including both the prompt and the story. For example:

- meta-llama/Meta-Llama-3-8B-Instruct: 8,192 tokens
- mistralai/Mistral-7B-Instruct-v0.2 : up to 32,000 tokens

However, in practice, the effective input capacity can be lower than the theoretical maximum, depending on the inference framework. In our experiments, we noticed that the actual usable input length was often significantly less than advertised. For instance, although Mistral-7B-Instruct claims to support up to 32k tokens, we found that its usable input on Hugging Face Inference Endpoints was much lower. With further research, we realized that this discrepancy likely stems from infrastructure limitations, memory constraints, or restrictions imposed by the hosting environment or API wrappers.

To estimate token usage effectively, we followed a general rule of thumb adapted from OpenAI’s tokenization guidelines: 75 English words $\approx$ 100 tokens

This rule helped us plan how much text we could safely pass to a model. Importantly, both the input and the model’s generated output must fit within the total context window. For example, if a model has an 8k token limit and the input already consumes 7,900 tokens, there is almost no space left for the output. This tradeoff becomes even more critical when prompts are used—since instructions, examples, or formatting requirements also consume tokens and must be accounted for in the total.

These constraints significantly influenced our summarization strategy and led us to adopt techniques like chunking (described in the following section), which allowed us to process longer texts while remaining within token limits.

Chunking for Long Inputs

To handle stories that exceed the token limits of the models, we implemented a chunking strategy. This involved splitting each story into smaller, overlapping segments that fit within the model’s maximum input length. We then generated a partial summary for each chunk independently. After obtaining these chunk-level summaries, we concatenated them into a single text and fed this combined summary back into the model with a prompt requesting a high-level, coherent summary of the entire story. If the concatenated summaries still exceeded the model’s token limit, we applied the same chunking and summarization process recursively, breaking the text into smaller parts until the final summary fit within the allowed input size. While this approach allowed us to process longer texts without truncation, it also had some drawbacks:

- **Loss of long-term coherence:** When a character (e.g., a princess) is introduced in one chunk and referred to by pronoun (e.g., “her”) in a later chunk, the model cannot resolve these references across chunks.

- **Repetition and redundancy:** Overlapping chunks can lead to duplicate content in summaries.
- **Computation cost:** Every model call consumes memory and compute. Even though we used free models, API rate limits (e.g., calls per second or per day) restricted throughput.

Extracting Patterns from Summaries

To identify narrative motifs, we initially analyzed the generated summaries. However, semantic variation posed a challenge: similar concepts (e.g., a journey, a magical item) were expressed through synonyms (e.g., “magic ring” vs. “enchanted band”), culturally specific symbols, or abstract descriptions. This variability made rule-based pattern extraction impractical without advanced semantic analysis. Consequently, we shifted to using LLMs to identify motifs directly.

Combining Summarization and Pattern Extraction

Our initial pipeline involved summarizing each folktale into a condensed text, then separately querying the LLM to identify recurring narrative patterns or motifs from these summaries. However, this method doubled the number of model calls and introduced additional risks of losing nuance during the intermediate summarization stage.

To address these challenges, we revised our approach to directly extract motifs from the raw story text using proper prompts. This combined method leverages the LLM’s ability to perform complex reasoning and pattern recognition in a single pass, reducing computational overhead and potential error accumulation. By skipping the separate summarization step, we preserve more of the original narrative context, improving the quality and coherence of the extracted motifs. Furthermore, this integrated strategy allowed us to utilize more powerful LLMs, such as Google Gemini, which is crucial for handling the complexity of motif

extraction directly from longer texts. Overall, this approach simplifies the workflow, reduces the number of API calls, and enhances the consistency and richness of the extracted narrative patterns.

In the next section, we explain the details of this method.

2.1.2 Direct Motif Extraction

To extract motifs directly from the stories, we used the Google API, to query the `gemini-2-flash` model.

The final prompt used for motif extraction is shown in Figure A.3 in the appendix. Developing an effective prompt required several rounds of experimentation, during which we gradually refined our initial versions to achieve outputs that were both usable and consistent.

A well-designed prompt needed to do several things simultaneously. First, it had to be sufficiently informative so that the model clearly understood that it was being asked to identify narrative motifs. At the same time, the prompt had to discourage the model from generating any irrelevant or unwanted content, such as full sentences, explanations, or free-form summaries. Since the output would later be used in pattern comparison and clustering, it was essential that motifs were extracted in a minimal, structured format without superfluous words. To help guide the model toward the desired behavior, we included a few illustrative examples within the prompt. This few-shot prompting strategy encouraged the model to mirror the format we wanted—namely, short, standalone phrases rather than full narrative descriptions.

Through our tests, we encountered a few recurring issues. One involved the model’s tendency to use slashes when expressing synonyms, such as “escape/run away”. This made it harder to standardize and compare motifs, so we explicitly instructed the model not to use such notation. Another problem was overgeneralization. For instance, in the case of the Cinderella story, we expected motifs like

“The prince marries the poor girl”, but the model sometimes returned only the word “marriage” which lacked the specificity needed to retain narrative meaning. To address this, we added a final instruction in the prompt encouraging the model to add slight, story-specific details for clarity. This allowed us to retain enough semantic information to make the motifs meaningful while still keeping them general enough to support comparison across different stories.

In the post-processing stage, we took the raw output generated by the language model and extracted the individual motifs. This involved cleaning the output to remove any unwanted characters, formatting inconsistencies, or artifacts introduced by the model. The cleaned motifs were then structured into a standardized list format, with each list corresponding to a single story.

2.1.3 Similarity analysis on motifs

To analyze the similarity between motifs, we first converted each motif into a numerical vector, or embedding, using sentence transformer models. We experimented with three different models: sentence-t5-base, sentence-t5-large, and roberta-large. These embeddings capture the semantic meaning of each motif in a high-dimensional space, where motifs with similar meanings are located closer together. To quantify similarity between two motifs, we calculated the cosine similarity between their embeddings.

For easier analysis, we grouped semantically similar motifs into clusters, treating each cluster as a single motif category. This clustering reduced noise caused by minor variations in phrasing and lowered the number of unique motifs to consider. For clustering we used agglomerative clustering, a hierarchical method that progressively merges motifs based on their embedding similarity, building a tree of clusters.

Agglomerative clustering suited our data well because it does not require specifying the number of clusters in advance—unlike k-means, which needs a pre-

set number that is hard to estimate for a large dataset. Additionally, it captures complex relationships better than k-means (which assumes spherical clusters) or DBSCAN (which may miss clusters with varying densities). The hierarchical structure also allows us to set a similarity threshold that balances grouping meaningful clusters while preserving subtle differences between motifs.

After clustering, we computed the centroid (average embedding) of each cluster and selected the motif closest to this centroid as the cluster’s representative. This approach reduced the original 1222 extracted motifs to 216 clusters, making it easier to identify and study recurring narrative patterns.

With the embeddings and clusters, we performed a range of analyses to explore the relationships between motifs, identify dominant themes, and investigate the distribution of motifs across the stories.

2.2 Results

2.2.1 Relationship of motifs and ATU indices

In this section, we first employ qualitative analysis to find the possible relationships between the extracted motifs and their corresponding ATU indices, aiming to uncover any visible patterns or groupings. Following this, we apply quantitative methods to evaluate the effectiveness of our motif extraction and representation approach in capturing narrative structures relevant to ATU classification.

Qualitative Analysis:

Figure 2.1 visualizes motifs corresponding to 10 randomly selected ATU indices, with each color representing a different ATU category. To project these high-dimensional motif embeddings into a two-dimensional space for easier interpretation, we applied a combination of Principal Component Analysis (PCA) followed by t-Distributed Stochastic Neighbor Embedding (t-SNE). (The necessity of

using PCA prior to t-SNE and its benefits are explained in the Appendix.)

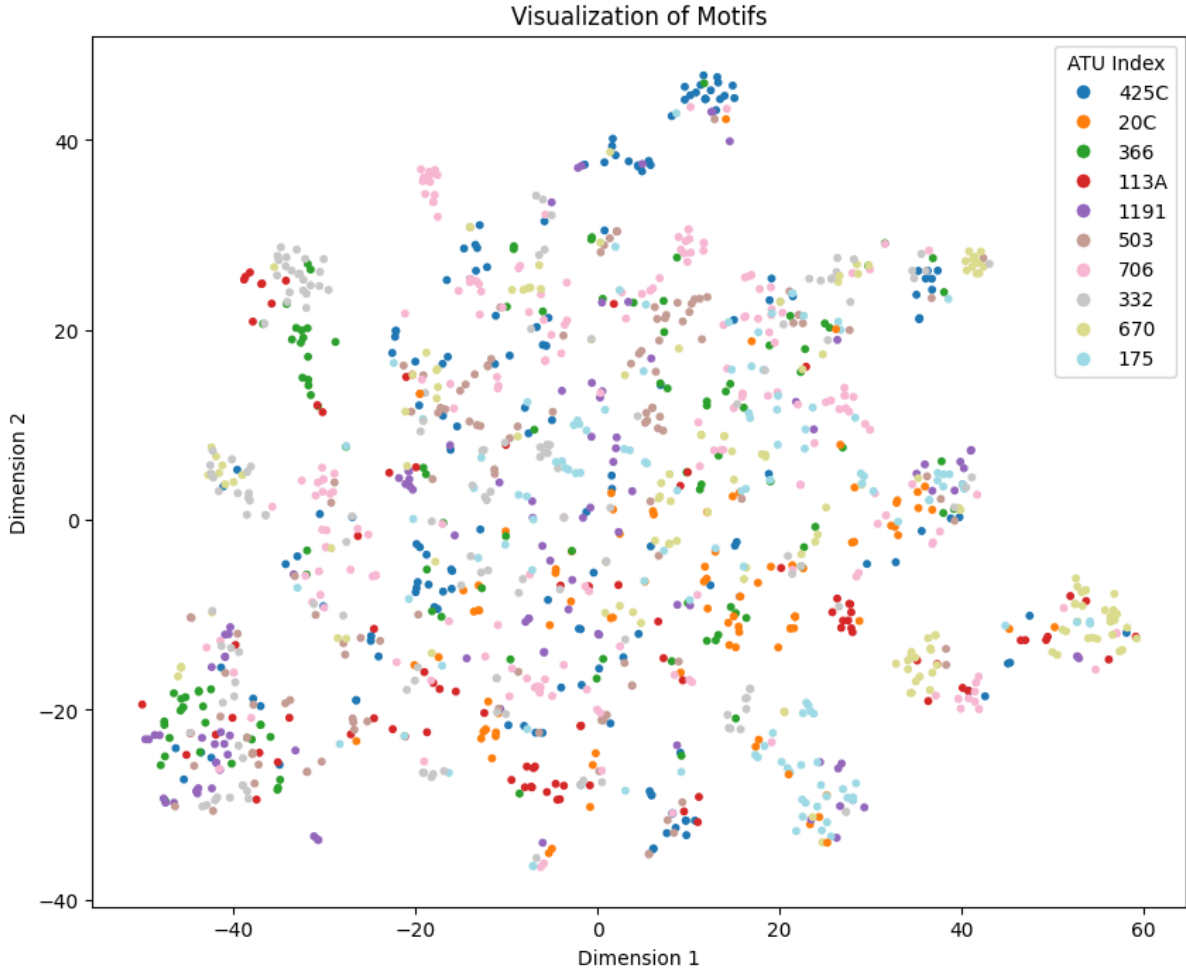


Figure 2.1: Visualization of motifs using the direct motif extraction approach. Each dot represents a single motif, and each color represents a different ATU category.

At first glance, the visualization of individual motifs showed no clear or distinct clustering by ATU index. However, this might be due to a rich motif extraction across ATU indices. This suggest that single motifs alone may not carry enough semantic information to directly reflect the complex ATU classification, which is expected given the richness and variability of narrative elements in folktales.

In contrast, Figure 2.2 presents the same 10 ATU indices that we visualized before using direct motif extraction and clustering, but this time each story is represented by the mean vector of all its motif embeddings. By averaging the embeddings of all motifs within a story, we derived a single vector that encapsulates the

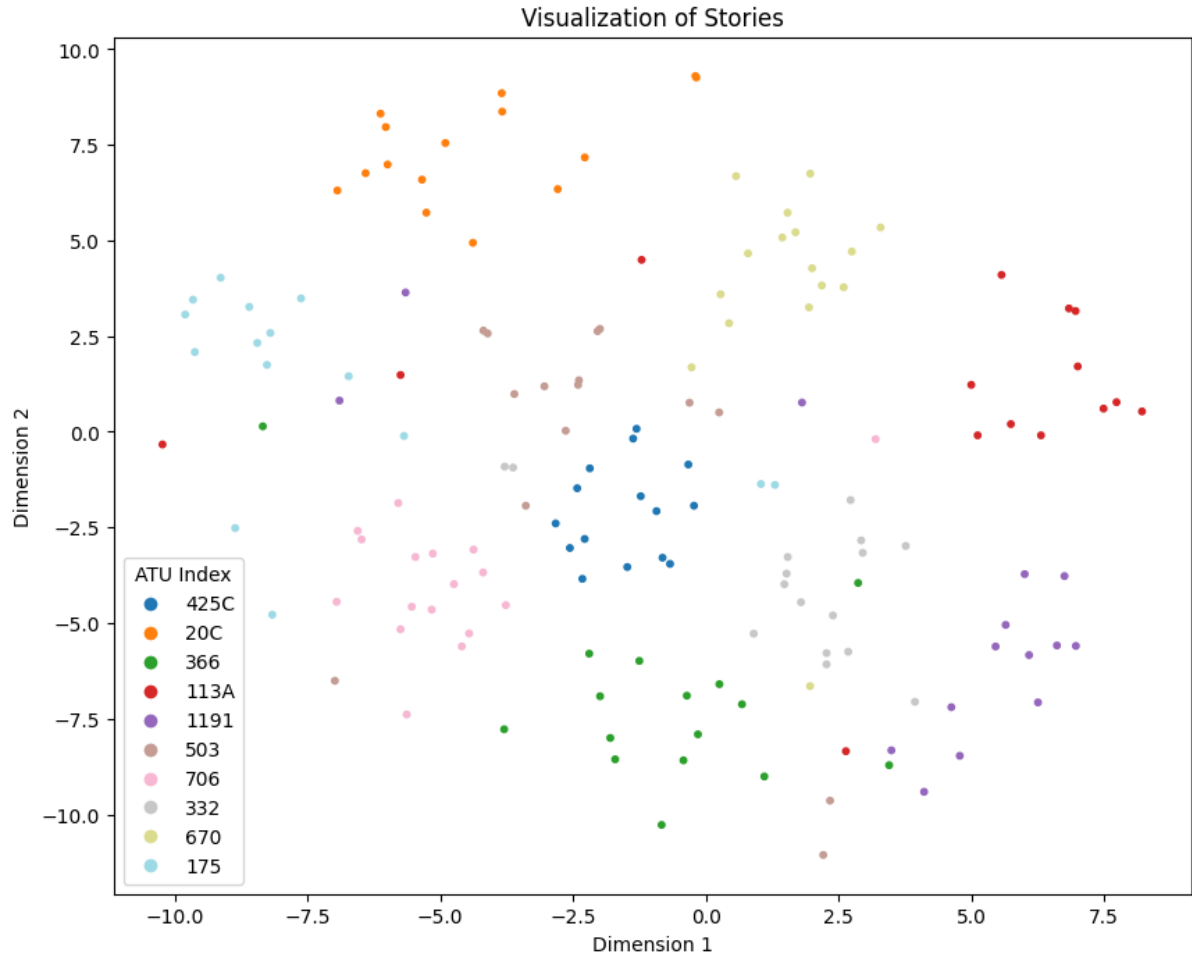


Figure 2.2: Visualization of stories. Each dot represents a story (calculated as the average of its motifs), and each color represents a different ATU category.

overall semantic profile of that story’s motifs combined. When these story-level vectors were visualized in 2D space, distinct clusters emerged, showing distinct groupings of stories that aligned with their respective ATU index. This not only validated the effectiveness of our direct motif extraction method but also shows that the classification of a story into an ATU index is not determined by a single motif but by the concurrence of multiple motifs. The presence of certain motifs, in combination, defines the story’s category, reinforcing the notion that the structure and meaning of stories are shaped by the interplay of various narrative elements rather than by isolated motifs.

Quantitative Analysis:

To quantitatively assess the effectiveness of our direct motif extraction method and embedding approach, we performed a classification experiment using ATU indices as labels. First, we split the dataset into training and test sets, reserving 20% of the data for testing. We made sure the test set included a diverse range of ATU types to fairly evaluate the model’s generalization across different story categories.

For each story, we represented its narrative content by calculating the mean of all its motif embeddings, producing a single vector that summarizes the collective semantic information of the motifs within that story. Within the training set, we then computed the centroid for each ATU index by averaging the story vectors belonging to that category.

Classification was performed on the test set by assigning each story the ATU label of the nearest centroid, based on cosine similarity between embeddings. This simple classifier allowed us to directly measure how well the motif embeddings captured the semantic distinctions between ATU categories.

Recognizing that accurate centroid estimation requires sufficient data per category, we initially filtered out ATU indices with fewer than 15 stories. Using this filtered dataset, we compared classification results across three different sentence embedding models. As expected, models with higher-quality semantic representations achieved better accuracy. Table 2.1 shows the classification results on the test data for these three models, and Figure 2.3 presents the confusion matrix for the model with the highest accuracy(sentence-t5-large)

Model	sentence-t5-large	sentence-t5-base	roberta-large
Test accuracy	88.64	84.09	76.14

Table 2.1: Test accuracy of the classification on the filtered dataset with a minimum threshold of 15 stories per ATU index, using different sentence embedding models.

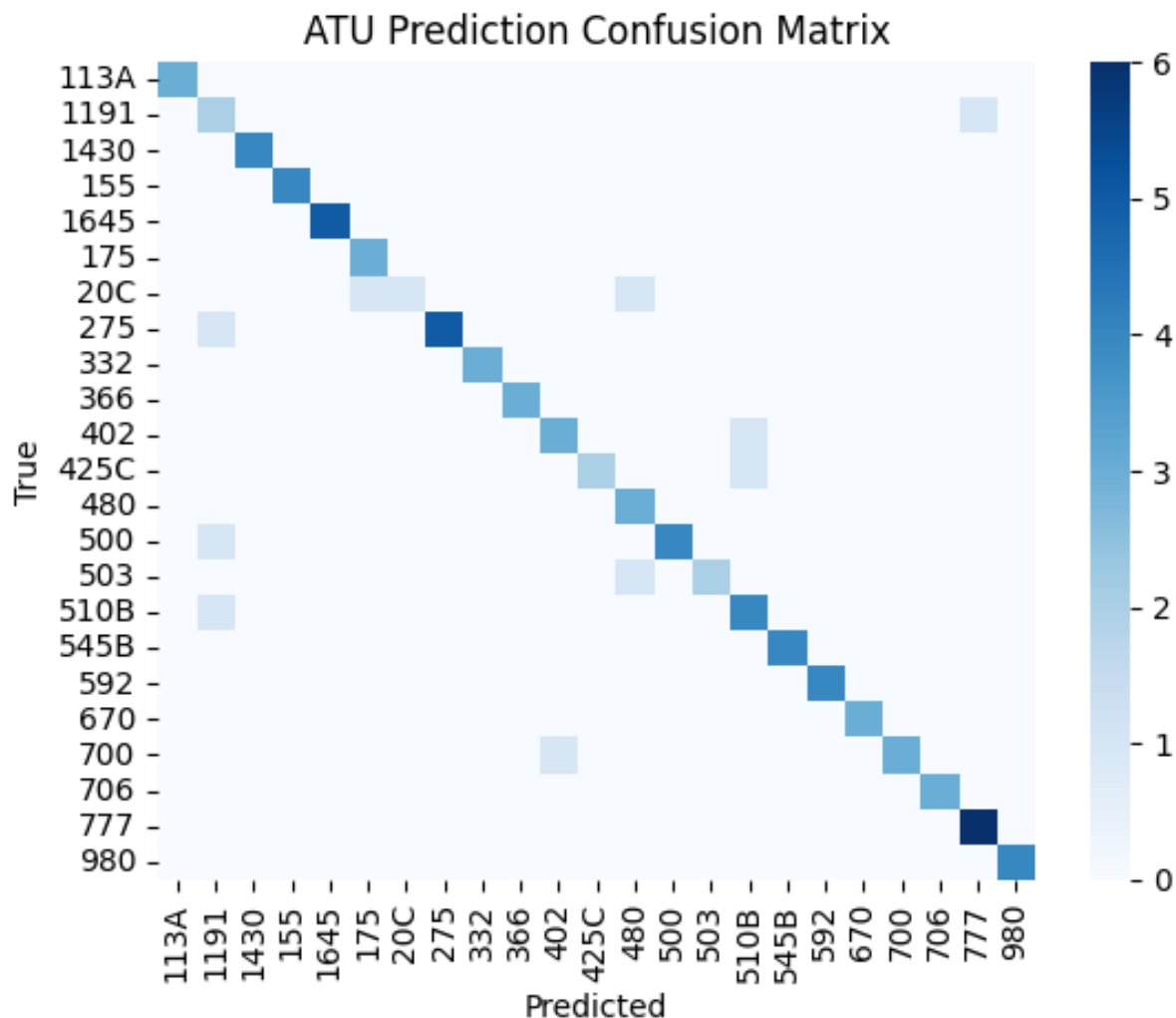


Figure 2.3: Confusion matrix for performance of the sentence-t5-large model on the test dataset, showing the distribution of predicted versus actual ATU indexes.

To further investigate the effect of the minimum number of stories per ATU index, we relaxed the filtering threshold to include ATU indices with at least 10 stories and repeated the experiment. Although we observed a clear drop in classification accuracy, the results remained significantly better than random guessing. This demonstrates that even with fewer stories per ATU category, the motif-based embeddings still retain meaningful narrative information that supports effective classification.

Model	sentence-t5-large	sentence-t5-base	roberta-large
Test accuracy	77.18	72.01	67.79

Table 2.2: Test accuracy of the classification on the filtered dataset with a minimum threshold of 10 stories per ATU index, using different sentence embedding models.

2.2.2 Most common motifs

To identify the most and least common motifs across different ATU indices obtained by direct motif extraction, we measured the number of distinct ATU indices each motif cluster appeared in. (Here, by “motif” we refer to the representative of each motif cluster, as described in Section 2.1.3.) Motifs that occur across a wide range of ATU indices represent recurring narrative elements that are shared among various folktale types. In contrast, motifs that appear in only a few indices tend to reflect more specialized or unique narrative features.

Table 2.3 presents the five most common motifs, along with the number of distinct ATU indices in which each appears. All of the five least common motifs appear in only a single ATU category. These are shown in Table 2.4, along with their associated ATU index and the title of the corresponding folktale type.

motif	number of ATU indices
Deception is used to trick another	117
Transgression results in punishment	101
Animal offers aid	73
Escape from danger	71
Reward received for kindness	69

Table 2.3: The five most common motifs and the number of distinct ATU indexes in which each appears (direct motif extraction).

2.2.3 Regional Patterns in Motifs

As explained in Section 2.1, the provenance column in our dataset refers to the origin of each tale. This could be the individual who recorded it, the region it came from, or the collection in which it was originally published. Initially, the

Motif	ATU Index	ATU Title
Decision to eliminate elders	981	Realistic Tales: Other Realistic Tales
Magical hair as access	310	Tales of Magic: Supernatural Adversaries
Evil is acknowledged as less harmful	1353	Anecdotes and Jokes: Stories about Married Couples
Superior knowledge proves useless	105	Animal Tales: Wild Animals and Domestic Animals
Race determines winner	275	Animal Tales: Other Animals and Objects

Table 2.4: The five least common motifs, their ATU index, and category name.

dataset included 379 unique provenance values.

We reviewed each of these values manually. For entries that referred to a person or a collection, we consulted Wikipedia to determine whether the associated story could be clearly linked to a specific region. If so, we replaced the provenance with the corresponding region.

In some cases, we found that a book or collection contained folktales from multiple origins that had simply been translated into another language. These entries were excluded, as they could not reliably represent a single geographical region. For older and well-known collections such as *The 1001 Nights* and *Aesop's Fables*, although their stories originate from particular historical areas, assigning them to a specific modern-day region is difficult. However, due to their cultural significance and authenticity, we chose to keep these collections in our analysis and use the name of collection as its region.

Finally, we removed all regions with fewer than five associated stories. After this filtering process, 27 distinct regions remained.

To explore regional patterns in motif distribution, we identified motifs that appeared across the widest and narrowest set of regions. Table 2.5 presents the five most widely distributed motifs—those occurring in the greatest number of

regions- and Table 2.6 presents five motifs that appeared in only one region.

motif	number of different regions
Deception is used to trick another	27
Punishment inflicted	24
Animal as helper	23
Reward for kindness	23
Disguise reveals true identity	23

Table 2.5: Five most widely distributed motifs and the number of distinct regions they appear in.

motif	Region
Houses as protection	England
Friendship tested by adversity	India
Hierarchical power structure revealed	India
Wealth and isolation	Scotland
Transformation of figures from old to new	Germany

Table 2.6: Five motifs that appear in only one region.

To further explore how motifs vary across folktales from different regions, we computed the mean embedding of all motifs associated with each region. These region-level embeddings serve as representative vectors that summarize the dominant narrative elements for each geographical area. We then applied t-SNE, to project these embeddings into two dimensions for visualization. The resulting plot (Figure 2.4) displays each region as a single dot.

Although our dataset lacks detailed metadata—such as the historical timeline of the stories, precise provenance, or information about translation and cross-cultural transmission—we can still see meaningful geographical patterns in the visualization.

One particular case is Aesop’s Fables, a collection traditionally attributed to Aesop, a storyteller believed to have lived in ancient Greece around 620–564 BCE. While Aesop’s stories are widely associated with European folklore, many of them exhibit striking similarities to Indian tales. For instance, several Aesopic fables closely parallel stories found in the Indian Jataka Tales and the Panchatantra,

both of which predate or overlap with the timeline attributed to Aesop. Scholars have noted over a dozen such narrative correspondences, often with thematic or structural alignment but variations in surface details (Berezkin and Duvakin 2016, G. 2019).

Our motif-based analysis reflects this connection: in Figure 2.4, the embedding representing Aesop’s Fables is positioned much closer to the Indian cluster than to other European regions. This spatial proximity suggests a strong overlap in motif usage and supports the hypothesis of shared narrative between Indian and Greek storytelling traditions.

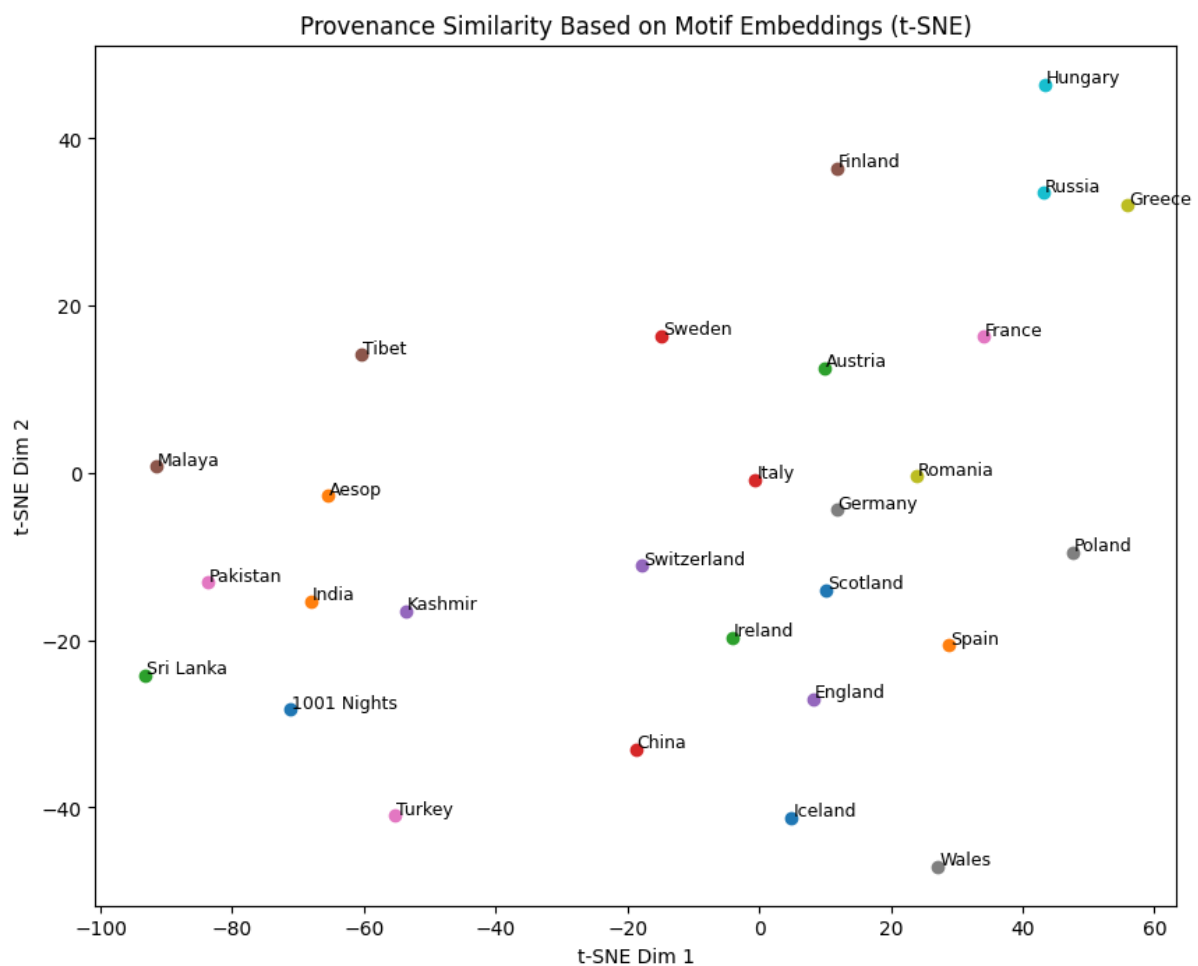


Figure 2.4: Visualization of regional narrative patterns. (Each point represents a geographical region, based on the mean embedding of motifs from stories attributed to that region)

Chapter 3

Approach 2

3.1 Main Character based Motif Extraction

One early experimental approach implemented using the gemini-1.5-flash model involved a multi-step prompting sequence designed to extract narrative patterns by focusing on the main character and key plot points. This method processed each tale through three distinct API calls:

- **Character Role Identification:** The first prompt asked the model to identify the social role of the main character at the story's beginning, constrained to a single word output.
- **Simple Plot Generation:** A second prompt then requested a plot summary using simple sentences starting with the identified character role, avoiding specific names.
- **Filtering for Specific Sentences:** The final prompt filtered the generated simple sentences, instructing the model to return only those specific to the individual story, excluding generic phrases.

The resulting output was a list of short sentences intended to represent key actions tied to the main character, examples including "Prince learned wisdom,"

”Servant escaped robbers,” and ”Woodcutter’s daughter outsmarted witch”. While this provided a structured plot representation, the multi-step process meant three API calls per story, thus increasing computational cost. Moreover, one could argue that extracting sentences only based on the main character of the story might not be effective for fully comprehending the important aspects of a story, e.g. in stories with multiple main characters.

Qualitative Analysis

To get an idea of the effectiveness of the method, we looked at the overlap of clustered sentences, which we will consider later as clustered motifs, and the ATU index of the corresponding stories. To do so we begin by representing each sentence as a TF-IDF vector, capturing the relative importance of words across the dataset. The obtained vectors are then projected into a lower-dimensional space to reduce noise and computational load. Next, using cosine similarity as the underlying metric, a t-SNE algorithm maps these vectors into two dimensions.

An example of clustered motifs obtained from stories of 10 randomly selected ATU indices, can be found in Figure 3.1. The clusters obtained by using the main character based method seem to significantly overlap with the ATU indices.

3.2 Keyword based Motif Extraction

Another experimental strategy, also implemented using the gemini-1.5-flash model, explored a two-step process focusing on keyword extraction and subsequent sentence generation based on those keywords. This method aimed to ground the pattern extraction in specific plot elements. The process for each tale involved:

- **Keyword Extraction:** The first prompt requested a list of five simple, common keywords relevant to the plot, excluding character names and strictly limiting the output to just the keywords.

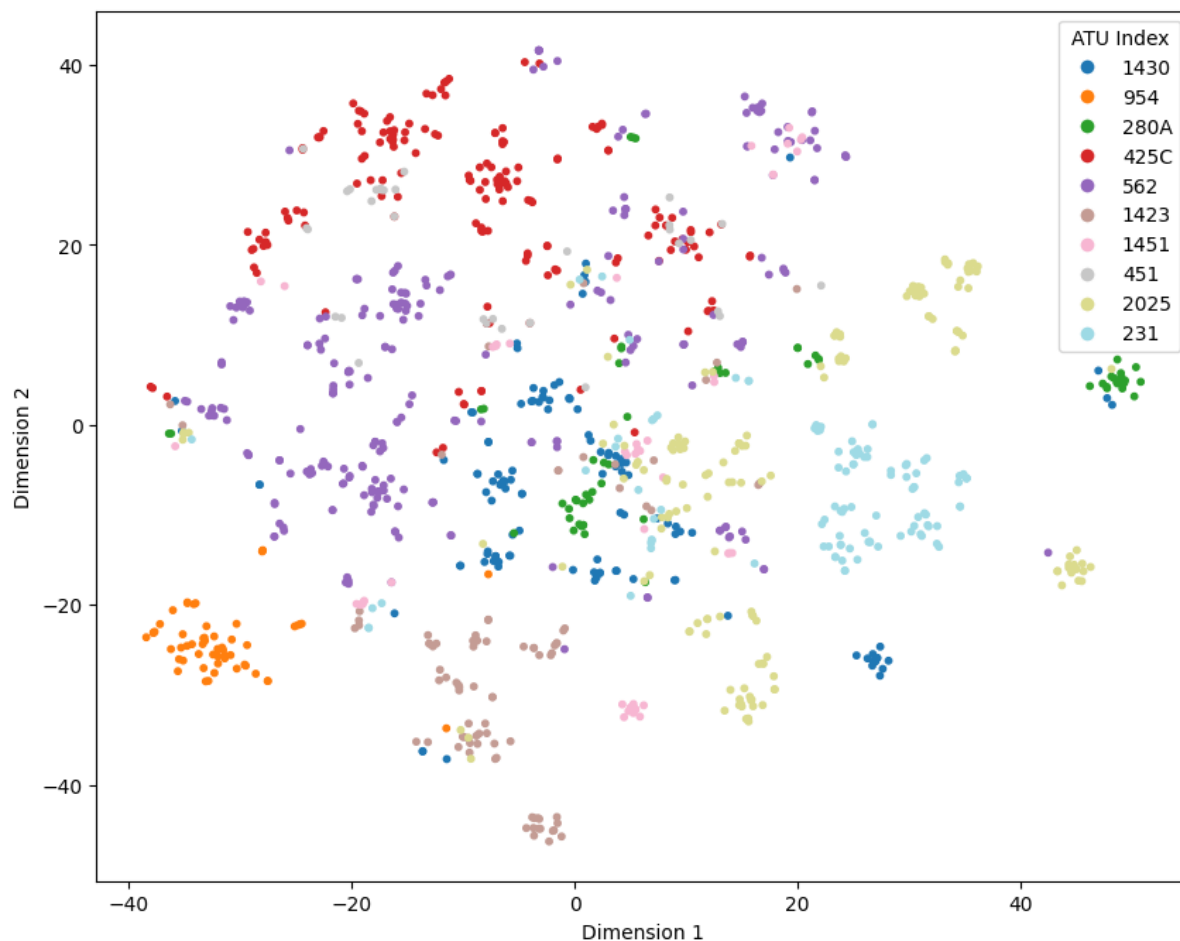


Figure 3.1: Visualization of motifs using the main character based approach. Each dot represents a single motif, and each color represents a different ATU category.

- **Keyword Relevance Sentences:** The second prompt used the story text and the extracted keywords to generate extremely short, concise sentences explaining the relevance of each keyword to the plot, using examples to guide the desired "Keyword: Description" format.

This approach yielded "Keyword: Sentence" pairs, such as "Stepmother: The stepmother mistreated Cinderella" and "Slipper: The slipper identified Cinderella". This method provided a structured, element-focused output and reduced the calls to two per story compared to the previous three-step method. However, it still involved multiple calls and relied heavily on the quality and relevance of the initial keyword extraction step influencing the final patterns.

Qualitative Analysis

Once again we used TF-IDF vectorization combined with t-SNE dimensionality reduction, guided by cosine similarity, to visualize and explore the semantic structure of the produced sentences.

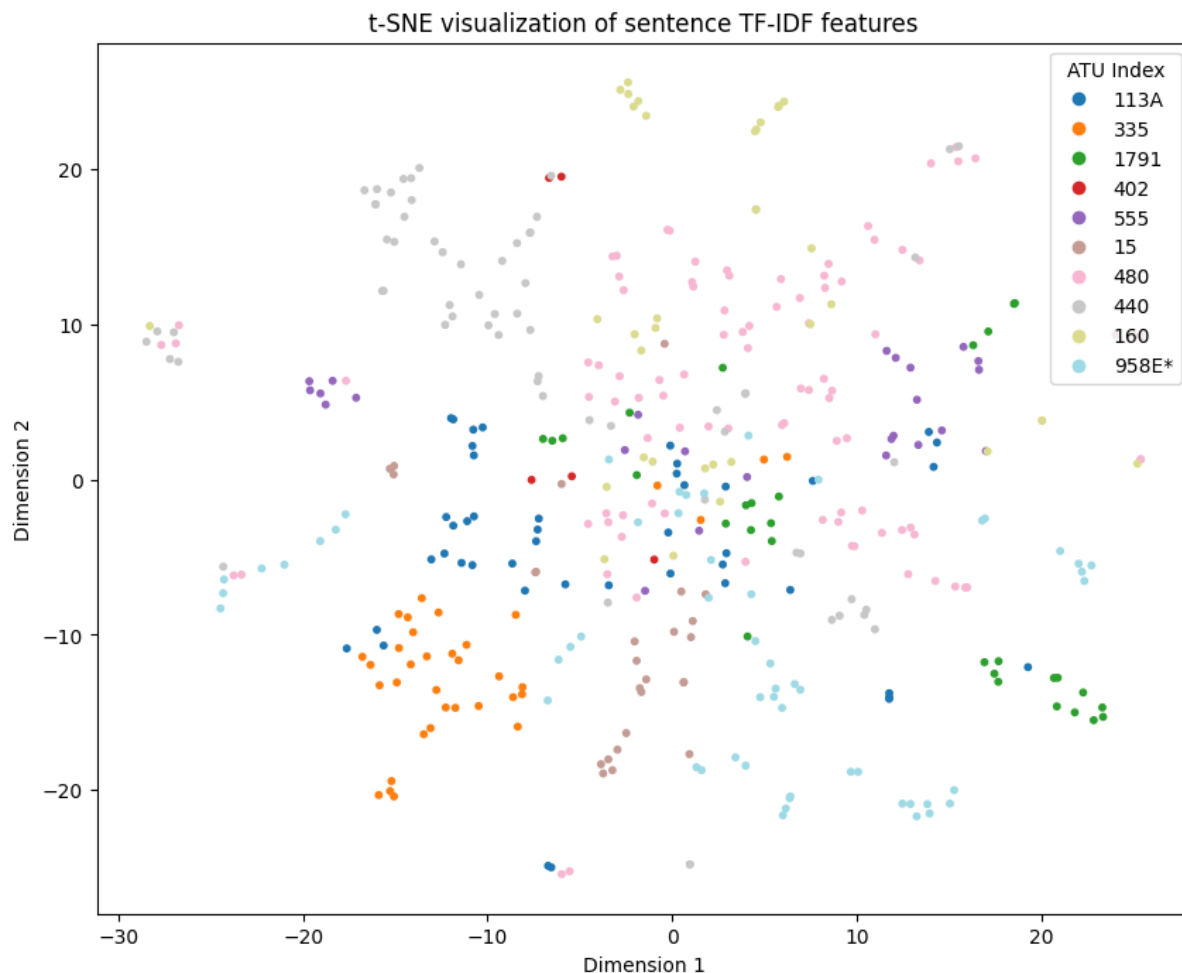


Figure 3.2: Visualization of motifs using the keyword based approach. Each dot represents a single motif, and each color represents a different ATU category.

In Figure 3.2, we can see the resulting clusters of 10 random ATU indices. The clusters resemble the ATU indices, but we also have clearly mixed clusters.

3.2.1 Extracted Motifs

From the representation of the sentences obtained by TF-IDF vectorization, t-SNE dimensionality reduction and cosine similarity we separated the results from the

keyword based method into cluster using Kmeans (number of clusters=130). To extract motifs from our results, we used our LLM, gemini-1.5-flash, to find a common motif among sentences of the same cluster. The used prompt can be found in Appendix A.4. Some of the extracted groups had to be manually filtered out, since they were too general (e.g. results like "Many of these sentences describe a key object, character, or event that is central to the plot."). However, after this sorting we were left with an interesting list of motifs across multiple ATU indices. The first entries of this list can be found in Table 3.1.

Motif	Number of ATU indices
The king is a central figure whose actions, decisions, or fate are influenced by various events and characters.	64
The narratives frequently depict the negative consequences of greed , often resulting in misfortune or death for the greedy characters , sometimes involving older men.	43
Many of these narratives involve transformations, magical objects, or challenges overcome through cleverness or trickery .	32
Many of these narratives involve betrayal, deception, and/or tests of character .	30
Many of these sentences describe a man's journey , often involving challenges, tests, and transformations, sometimes leading to wealth or misfortune.	29
Gold is a recurring motif representing wealth, reward, or a source of conflict and greed, often discovered unexpectedly or obtained through challenging tasks.	29
Money plays a central role in these stories, often driving the plot through theft, discovery, or its use as a reward, bet, or source of conflict .	28
Many of the stories center on the complex relationship between a husband and wife , often involving deception, betrayal, or conflict.	25
Houses serve as significant settings and locations for various events and actions in the stories.	25
A prince , often undergoing transformations or disguises, seeks a princess's hand , overcoming obstacles and challenges along the way.	24

Table 3.1: Motifs and their Number of ATU indices (keyword based).

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Appendix A

Appendix

Selection and Use of Google Gemini Models for Motif Extraction

Following the initial exploration with other models and the decision to move towards direct information extraction from raw text, we focused on Google Gemini models for the experimental phase of developing the motif extraction prompts. Gemini models were chosen due to their capabilities in following instructions, processing complex text, and being accessible via API, aligning with our criteria for free accessibility and efficiency.

Using PCA prior to T-SNE:

T-SNE was chosen for its ability to preserve local structures in high-dimensional data, making it particularly suitable for our case. By emphasizing local relationships, t-SNE ensures that motifs with similar semantic or thematic characteristics appear closer together in the 2D projection, facilitating the identification of patterns or groupings that align with ATU classifications. However, t-SNE has limitations when applied directly to very high-dimensional data, such as our motif embeddings. Specifically, t-SNE's computational complexity increases significantly with dimensionality, and it can struggle to converge or produce meaningful results if the input space is too large or noisy. This is where PCA plays a crucial

role. Before applying t-SNE, we first reduced the dimensionality of the embeddings using PCA, which compresses the original high-dimensional data into a smaller number of principal components while retaining most of the variance and important structural information. This step simplifies the input for t-SNE, improves computational efficiency, and reduces noise, enabling t-SNE to better capture meaningful local patterns in a manageable lower-dimensional space.

ATU hierarchy

Table A.1 shows the ATU index hierarchy,

ATU-Index Range	Category	Subcategories
1-299	Animal Tales	Wild animals, Wild and domestic animals, Animals and humans, Domestic animals, Others
300-749	TALES OF MAGIC	Supernatural Adversaries, Supernatural or Enchanted Wife (Husband) or Other Relative, Supernatural Tasks, Supernatural Helpers, Magic Objects, Supernatural Power or Knowledge, Others
750-849	RELIGIOUS TALES	God Rewards and Punishes, The Truth Comes to Light, Heaven, The Devil, Others
850-999	REALISTIC TALES	The Man Marries the Princess, The Woman Marries the Prince, Proofs of Fidelity and Innocence, The Obstinate Wife Learns to Obey, Good Precepts, Clever Acts and Words, Tales of Fate, Robbers and Murderers, Others
1000-1199	TALES OF THE STUPID OGRE (GIANT, DEVIL)	Labor Contract, Partnership between Man and Ogre, Contest between Man and Ogre, Man Kills (Injures) Ogre, Ogre Frightened by Man, Man Outwits the Devil, Souls Saved from the Devil
1200-1999	ANECDOTES AND JOKES	Stories about a Fool, Stories about Married Couples, Stories about a Woman, Stories about a Man, Jokes about Clergymen and Religious Figures, Anecdotes about Other Groups of People, Tall Tales
2000-2399	FORMULA TALES	Cumulative Tales, Catch Tales, Others

Table A.1: The ATU index ranges, categories, and subcategories from the ATU hierarchy.

LLM Prompts

Main Character based Motif Extraction

What is the role in society of the main character at the beginning of the story: {story}? Return only one word.
Give the plot of story in simple sentences of 3 to 5 words. Start the sentences with the word {characterRole} and do not give names.
Return only the sentences that are specific to a certain story and not global formulations: {plot} Do not return anything else beside the sentences.

Figure A.1: Prompts used in keyword based extraction.

Keyword based Motif Extraction

Give me a list of five keywords that are relevant to the plot of the following story (excluding the characters): {text}. The keywords should be simple, common words. Give nothing else, only the keywords.
{keywords} this is a list of five keywords relevant this story: {story}. Can you give for each one a simple sentence explaining its relevance to the plot. The sentences should be extremely short and concise. Examples: keyword betrayal, the step mother betrayed the step daughter. ; apple: the princess ate the poisoned apple. ; etc. Do not return anything else.

Figure A.2: Prompts used in the keyword based extraction.

Direct Motif Extraction

Given the following folktale, extract key motifs as a list of short, general phrases (e.g., ‘magical helper aids,’ ‘journey undertaken,’ ‘reward earned’). Avoid using synonyms with backslashes (e.g., ‘scarcity strikes,’ not ‘famine/scarcity’). Add slight detail for clarity, but keep phrases general and applicable across stories, avoiding specific names or objects. Folktale: {story} Motifs:

Figure A.3: The final prompt used in the experiments.

Extracting motifs from clusters (keyword based approach)

Given these sentences that share a similarity: {keywordsWithDescriptions}. Provide one concise sentence that describes the similar plot or return None if the plots are too different. Return nothing else.

Figure A.4: Prompt used for generating motifs from the keyword based approach and clustering.